LoRaWAN based Water Level Monitoring Protocol Document

Uplink Messages: (Device -> Thingsboard)

1. Water Level Data Payload:
   A10000000D1A850000000000
   - A1 -> Type of payload
   - 0000000D -> Packet Number ($0000000D_{16} = 13_{10}$)
   - 1A85 -> Water Level ($1A85_{16} = 6789_{10}$ = 67.89%)
   - 00 -> Reserved_1
   - 00 -> Reserved_2
   - 00 -> Reserved_3
   - 00 -> Reserved_4
   - 00 -> Reserved_5

Note: The water level value is stored as an integer and must be divided by 100 to determine the actual percentage.

2. Current Device Configurations Retrieval ACK / Device Configuration Update ACK:
   A20000000D0E1000000000
   - A2 -> Type of payload
   - 0000000D -> Packet Number ($0000000D_{16} = 13_{10}$)
   - 0E10 -> Upload Interval (sec) ($0E10_{16} = 3600_{10}$)
   - 00 -> Reserved_1
   - 00 -> Reserved_2
   - 00 -> Reserved_3
   - 00 -> Reserved_4

Downlink Messages: (Thingsboard -> Device)

1. Update Device Configurations:
   B1070800000000
   - B1 -> Type of payload
   - 0708 -> Upload Interval (sec) ($0708_{16} = 1800_{10}$)
   - 00 -> Reserved_1
   - 00 -> Reserved_2
   - 00 -> Reserved_3
   - 00 -> Reserved_4

2. Fetch Device Configurations:
   B2
   - B2 -> Type of payload